Circular Queue Using Array – C Program

```
#include <stdio.h>
#define SIZE 5

int queue[SIZE];
int front = -1, rear = -1;

void enqueue()
{
    int value;

    if ((rear + 1) % SIZE == front)
    {
        printf("Queue is full");
        return;
    }

    printf("Enter value: ");
    scanf("%d", &value);

    if (front == -1)
        front = 0;

    rear = (rear + 1) % SIZE;
    queue[rear] = value;

    printf("Element inserted");
}

void dequeue()
{
    if (front == -1)
    {
        printf("Queue is empty");
        return;
    }

    printf("Deleted element: %d", queue[front]);

    if (front == rear)
    {
        front = -1;
        rear = -1;
    }
    else
        front = (front + 1) % SIZE;
}

void display()
{
    int i;

    if (front == -1)
    {
        printf("Queue is empty");
        return;
    }

    printf("Queue elements: ");
    i = front;

    while (1)
    {
        printf("%d ", queue[i]);

        if (i == rear)
            break;

        i = (i + 1) % SIZE;
    }
}
int main()
```

```
{
    int choice;

    do
    {
        printf("\n\n1.Enqueue  2.Dequeue  3.Display  4.Exit");
        printf("\nEnter choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1: enqueue(); break;
            case 2: dequeue(); break;
            case 3: display(); break;
            case 4: printf("Exit"); break;
            default: printf("Invalid choice");
        }
    } while (choice != 4);

    return 0;
}
```